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The Influence of Sport Representation and Attitudes Toward Strength Training on Neuromuscular Performance Profiles in University Athletes: Part IIMale Athletes

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Abstract

McClean, ZJ, da Silva Torres, R, Herzog, W, Pasanen, K, Lun, V, and Jordan, MJ. The influence of sport representation and attitudes toward strength training on neuromuscular performance profiles in university athletes: Part I—Male athletes. *J Strength Cond Res* XX(X): 000–000, 2025—Strength, power, and plyometric testing are essential to evaluate neuromuscular performance in athletes. However, this approach creates datasets with numerous outcome measures that can lead to challenges for interpretation and establishing relevant performance benchmarks for preseason testing and performance-readiness after injury. The idea that athlete performance profiles exist within a larger population has been suggested, but limited research has explored this concept or suggested methodologies for delineating relevant profiles. Exploring the existence of neuromuscular performance profiles in university athletes while accounting for the influence of the sport environment and psychosocial factors, such as attitudes toward strength training, may support more athlete-specific neuromuscular benchmarks. Healthy male university athletes ($n = 272$) from 5 sports completed a comprehensive neuromuscular performance testing battery and a questionnaire that included assessment of attitudes toward strength training. Unsupervised machine learning applied to the body weight–normalized neuromuscular performance dataset, along with Fisher's exact tests, was used to examine differences in attitudes toward strength training across clusters (alpha = 0.05). Five profiles were identified, including a high strength/high power/braking-dominant jump strategy cluster with a large ice hockey representation and a high strength/high power/fast jump strategy cluster consisting mostly of field-sport athletes. Differences in attitudes toward training were noted across profiles ($p < 0.05$); for instance, athletes in a low-strength/low-power profile tended to prefer training in a more private training environment ($p = 0.023$). These results may help inform neuromuscular performance benchmarks in male university athletes, while the psychosocial characteristics of these profiles may provide insight into increasing strength training engagement in this population.

Key Words: athlete profiling, countermovement jump testing, sport injury, sport performance, unsupervised machine learning

Introduction

Lower limb neuromuscular performance testing that uses a biomechanical force assessment to evaluate maximal strength, power, and plyometric capacities is a cornerstone in the physical preparation process of athletes (12,24) and can be used to guide postinjury rehabilitation (18) and return-to-sport decision making (12). However, this approach raises challenges for benchmarking and test interpretation due to the multitude of outcome measures describing eccentric and concentric strength capacities along with stretch-shortening cycle mechanics in jumping and hopping tasks. For instance, basing the interpretation of these datasets on group-level descriptive statistics across isolated metrics may fail to characterize the global neuromuscular performance patterns exhibited by an individual athlete (5). In addition, there is limited evidence supporting the use of cut-off values from a normative distribution for performance benchmarking and establishing performance-readiness after injury (5,12), which further limits the practical insights gained from this analysis method. Finally, the relative importance of each testing outcome metric for describing neuromuscular performance capabilities is unclear (5) and could impact the way that an athlete's neuromuscular performance profile–specific key performance indicators are identified.

Unsupervised machine learning (ML) is a promising method to address these gaps by capturing the underlying patterns in multidimensional datasets using a weighted combination of metrics to identify subpopulations of athletes belonging to a shared performance profile (5,8). The notion that elite athletes can be profiled based on a shared pattern of physiological characteristics has been proposed elsewhere as an *athlete phenotype* (35), but to date, there is limited scientific evidence to support this idea, and the best statistical methods to establish athlete profiles are unclear. To the best of our knowledge, no other study has applied unsupervised ML to a multifaceted biomechanical force testing dataset to explore the existence of neuromuscular performance profiles in healthy male university athletes, despite potential

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applications in interpreting preseason baseline testing and establishing performance-readiness after injury. However, unsupervised ML has produced practically relevant outcomes in several other research settings, including identifying individuals likely to experience better and worse anterior cruciate ligament (ACL) surgical outcomes (21) and as part of a talent evaluation process in young swimmers (40).

In addition, the athlete's environment influences neuromuscular performance (24). For instance, repeated exposures to sport-specific training stimuli result in specific neuromuscular adaptations (10). Furthermore, psychosocial factors such as an athlete's perception of overall musculoskeletal function (assessed using the Sport Fitness Index (SFI), which is linked to sport injury risk (41) and biomechanical movement patterns (22) in university athletes) and attitudes toward strength training have been identified as potential barriers for engagement in heavy strength training among women (15). However, biomechanical/physiological neuromuscular performance assessments are rarely implemented concurrently with an evaluation of psychosocial factors, especially in a male university athlete population. Considering contextual factors within an exploration of neuromuscular performance profiles in healthy male university athletes could support a more individualized approach to augment engagement in heavy strength training and help guide practitioners in implementing profile-specific interventions to improve athletic performance and reestablish performance-readiness after injury.

Therefore, the purpose of this study was to (i) apply an unsupervised ML analysis to a biomechanical force testing dataset that characterized neuromuscular performance across maximal strength, power, and plyometric capacities in healthy male university athletes to test for the existence of distinct athlete profiles and (ii) compare contextual (sport representation) and psychosocial (attitudes toward strength training, and overall perception of limb function using the SFI) between neuromuscular performance profiles.

Methods

Experimental Approach to the Problem

As part of a multiyear longitudinal university athlete monitoring program, expansive neuromuscular performance testing was conducted with male university athletes at the start of the sports season in 2 consecutive years. Subjects were familiarized with the baseline muscle performance testing protocols, including double-leg CMJ testing with and without additional external load, single-leg CMJ testing, single-leg repeated hopping, and multijoint eccentric and concentric maximal strength tests (22). Before baseline testing, subjects completed a standardized and task-specific warm-up (22,25). After testing, subjects completed an online form that contained demographic information, the SFI (41), and an abbreviated version of the Attitudes Toward Strength Training and Injury Prevention Questionnaire (25).

Subjects

Healthy male university athletes (n = 218) from 5 team sports (soccer, basketball, volleyball, tackle football, ice hockey) were recruited using a cluster sampling approach. Athletes provided their written informed consent. Baseline testing was conducted over 2 consecutive years; 218 unique athletes participated in this study, and 54 athletes (25%) competed at the university sport level during consecutive testing seasons, resulting in a total sample size of 272 male athletes. Subjects were included in this study if they (a) were in the preseason phase of the off-season and preparing for competition in the upcoming university sport season; (b) had been cleared by an athletic therapist to perform double-leg and single-leg maximal jumping and maximal lower-body strength testing; and (c) had not sustained an ACL injury in the past 24 months. Subjects were excluded from this study if, for any reason, they were unable to perform maximal strength testing and/or maximal effort dynamic movements, such as CMJ. The protocol for this study was approved by the research institution's Conjoint Health Research Ethics Board (REB15-1094) and conducted in accordance with the Declaration of Helsinki (without registration).

Procedures

Countermovement Jump Testing. Biomechanical force assessments were conducted on a dual force plate system (Accupower, Advanced Measurement Technology Instruments, Boston, MA), sampling at 1000 Hz was used for all double-leg CMJ (CMJ, repetitions = 5), single-leg CMJ (CMJ_{SL}, repetitions = 3/leg), and double-leg CMJ with an extra external load (CMJ_{loaded}, extra load = 60% of subject's body mass, repetitions = 3) testing protocols. Using a strong verbal cue, subjects were instructed to maximize jump height while keeping their hands on their hips (6,16,18). For CMJ_{loaded} trials, subjects held the high handles of the hexagonal bar equal to 60% of their body weight.

The vertical ground reaction forces (Fz) for each force plate were measured and then analyzed using a custom-built computer program (MATLAB R2022a, Mathworks, Natwick, MA) (16). The subject's body mass was measured during a 2-second stationary standing period (16,17). The velocity of the body's center of mass (BCM) was used to delineate countermovement jump phases (16). The eccentric deceleration CMJ phase was defined as the period of time from when BCM achieved maximal negative (downward) velocity to when the BCM had a velocity of 0 m/s (16). The concentric CMJ phase was defined as the period of time from BCM = 0 m/s to jump toe-off (16). Jump height was calculated using the takeoff velocity (jump height = takeoff velocity^2/(2 g)), and the absolute system mechanical power was calculated continuously as the product of Fz and BCM velocity for the concentric and eccentric CMJ phases. Lower limb eccentric stiffness (LL_{STIFF}) was calculated as the change in Fz divided by the change in the downward displacement of the body center of mass during the eccentric deceleration phase (N/m), which was then body mass normalized for the analysis. Force at zero velocity was defined as the Fz value at the onset of the concentric phase. The percent difference between the maximal velocity obtained during the concentric phase (V_{max}) and the velocity at takeoff (TOV) was used to calculate the V_{max}-TOV difference.

Single-Leg Repeat Vertical Hop (RH_{SL}) Testing. On the above-mentioned force plate system, subjects performed 20 continuous single-leg repeat hops with their hands on their hips (RH_{SL}) under the cueing condition to maximize hop height while minimizing ground contact time. The maximal hop height was assumed to occur at the midpoint of the flight phase and was used to calculate the subsequent hop's landing velocity so that discrete hop movement phases could be calculated using the BCM velocity. The average hop performance over the middle 16 hops on each leg was used for analysis.

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Single-Leg Isokinetic Leg Extension Testing. Single-leg maximal isokinetic strength testing was performed on both the left and right leg using a multijoint robotic servomotor leg press device (Fv Seger, Treadmetrix, Park City, UT; sampling frequency = 200 Hz) and a standardized subject position (23). Before testing, a task-specific warm-up protocol was completed, which included 5 concentric-eccentric isokinetic repetitions (range of motion: 10–100 degrees of knee flexion, velocity = 0.2 m/s (33)) completed with increasing intensity (0, 50, 50, 70, 90% of the subject's perceived effort) and 3 unilateral isometric contractions on each leg at a 60-degree knee flexion angle. Following a 3–5-minute rest period, isokinetic maximal strength testing was performed using the same range of motion and velocity as the warm-up. During the 3 isokinetic repetitions, the instruction "push as hard as possible throughout the duration of the repetition" along with strong verbal cues of "push, push!" during concentric contractions, and "resist, resist!" during eccentric contractions were used (23). The maximal concentric (CON_{MVC}) and eccentric (ECC_{MVC}) force values achieved across all repetitions were obtained.

Statistical Analyses

Performance metrics with outlying datapoints more than 3 standard deviations away from the group mean were removed. Approximately 0.2% of all concentric strength, eccentric strength, CMJ, CMJ_{SL}, CMJ_{loaded}, and RH_{SL} values were removed as outliers. For CMJ, CMJ_{SL}, CMJ_{loaded}, and RH_{SL} testing, performance metrics were averaged across all the trial values included for analysis (9). For maximal concentric and eccentric muscle strength, the maximal force output obtained during left leg and right leg testing was used for analysis (22,23).

Questionnaires. The original 10-question SFI (41) assesses an athlete's perception of their overall musculoskeletal function and is associated with injury history (41), injury risk (41), and lower limb asymmetries (22). Each SFI question was numerically coded and summed to obtain a score from 0 to 100 where higher scores reflected better perception of limb function (41).

An abbreviated version of the Attitudes Toward Strength Training and Injury Prevention Questionnaire that only included the 14 non–sex-specific questions (25) was completed. Each question was answered on a 7-point Likert scale ("strongly agree" to "strongly disagree"). To understand more extreme attitudes toward each question, the 7-point Likert scale was categorized into 3 broader groups, where strongly agree and agree; slightly agree, neither, and slightly disagree; and strongly disagree and disagree responses were grouped together, respectively (25).

Data Processing. A matrix of biomechanical features (272 subjects × 39 features) was created. Overall, 1.5% of the total observations were missing, and, in all cases, missing data were due to (a) the subject's inability to grip the hexagonal bar during CMJ_{loaded} trials or (b) the subject declining to complete CMJ_{SL}, RH_{SL}, or maximal strength testing. For all relevant features, values were normalized to body mass on an athlete-by-athlete basis. Next, the Scikit-learn package (v. 0.20.4) was used to scale all features according to the interquartile range (RobustScaler function) and impute missing data based on the mean value of the 5 nearest neighbors in the dataset (KNNImputer function (37)).

Injury Surveillance. In collaboration with the university's athletic therapy department, injury surveillance was conducted to identify athletes who went on to experience a noncontact ACL injury in the sport season following that year's baseline testing. Injuries were recorded on a weekly basis using an electronic medical record system. Athletes who went on to experience a noncontact ACL injury were included for analysis. Noncontact ACL injury was defined following previous recommendations (31). It included scenarios where there was no direct contact with the athlete (i.e., sudden onset) and scenarios where the athlete was contacted by another athlete or an object at the time of injury, but the contact did not occur directly to the injured area. Exposure data were not recorded.

Statistical Analyses

Statistical analyses were conducted using the Python Language (v. 3.14). To determine the most appropriate clustering algorithm, cluster quality (i.e., Silhouette score, Davies Bouldin Score, and Calinski Harabasz Score) was computed for each combination of dimensionality reduction algorithm (i.e., Principal Component Analysis, t-distributed stochastic neighbor embedding, and Uniform Manifold Approximation and Projection for Dimension Reduction [UMAP)] and clustering algorithm (i.e., K-means, Hierarchical clustering analysis, Density-based Spatial Clustering of Applications with Noise, and Hierarchical Density-based Spatial Clustering of Applications with Noise). The combination of UMAP (20) (neighbors = 20, components = 10) and K-means clustering (32) (clusters = 5) produced the clustering outcome with the highest silhouette score. This result was then confirmed using the K-means "elbow method' (11). The neuralization-propagation (NEON) procedure (19) was used to provide insight into the relative contribution of each feature in the clustering assignment. The top 20 most important features for each cluster's centroid were obtained. Linear regression modeling was used to examine the effect of athlete age, experience level (i.e., number of years competing in elite sport), body mass, and SFI score on cluster assignment. Fisher's exact tests were conducted to examine differences in attitudes toward non–sport-specific training across athlete clusters. For all applicable statistics, an alpha level of 0.05 was used as the threshold to indicate significance.

Results

A K-means clustering approach identified 5 neuromuscular performance profiles in male university athletes (Silhouette score = 0.44, Davies Bouldin Score = 0.81, and Calinski Harabasz Score = 340.14). All athletes were included in a cluster (i.e., neuromuscular performance profile), and the characteristics of each cluster are presented in Table 1. A 2-dimensional visual representation of the clusters is shown in Figure 1.

Cluster 1: Low Strength and Power ($Cluster_{LOW}$)

$Cluster_{LOW}$ (n = 59, 21.7% of the total sample) was made up of tackle football (78.0% of the cluster), soccer (10.2% of the cluster), volleyball (6.8% of the cluster), basketball (3.4% of the cluster), and ice hockey (1.7% of the cluster) athletes. This was equivalent to 28.9% of the tackle football population, 13.7% of the soccer population, 13.8% of the volleyball population, 11.8% of the basketball population, and 4.4% of the ice hockey population. The top 20 features describing the biomechanical characteristics of this cluster are shown in Figure 2. Athletes in $Cluster_{LOW}$ had the lowest jump performance (jump height:

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Table 1

Neuromuscular characteristics of 5 male university athlete clusters measured across 5 testing batteries that evaluated maximal strength, power, and plyometric function.*

		Cluster centroid value, mean ± *SD*				
Testing protocol	**Outcome measure**	**$Cluster_{LOW}$**	**$Cluster_{SLOW}$**	**$Cluster_{AVG}$**	**$Cluster_{STRONG-BRAKE}$**	**$Cluster_{STRONG-FAST}$**
	Age (yrs)	20.5 ± 1.82	20.5 ± 1.9	20.6 ± 1.9	21.6 ± 1.9	20.2 ± 1.6
	Body mass (kg)	115.8 ± 25.9	84.4 ± 10.8	86.6 ± 11.8	85.8 ± 8.9	82.4 ± 10.0
Body mass CMJ	Jump height (cm)	29.3 ± 5.1	41.3 ± 3.5	37.1 ± 4.1	44.0 ± 5.3	41.3 ± 6.4
	Peak propulsive power (W/kg)	46.3 ± 5.5	57.1 ± 4.6	56.6 ± 4.8	59.8 ± 5.3	62.3 ± 6.9
	Rate of power development (W/kg/s)	205.4 ± 42.6	233.2 ± 32.9	292.0 ± 42.1	279.9 ± 34.4	377.3 ± 59.1
	Net eccentric phase impulse (Ns/kg)	1.08 ± 0.18	1.36 ± 0.17	1.18 ± 0.15	1.55 ± 0.15	1.3 ± 0.21
	Peak braking power (W/kg)	13.4 ± 3.2	17.7 ± 3.5	15.6 ± 3.0	23.4 ± 3.7	20.0 ± 5.5
	Concentric phase duration (s)	0.30 ± 0.04	0.31 ± 0.03	0.26 ± 0.02	0.28 ± 0.02	0.23 ± 0.02
	Eccentric phase duration (s)	0.21 ± 0.05	0.21 ± 0.03	0.18 ± 0.02	0.17 ± 0.02	0.14 ± 0.02
	Contraction time (s)	0.96 ± 0.14	0.98 ± 0.08	0.86 ± 0.06	0.90 ± 0.01	0.76 ± 0.08
	Whole body eccentric stiffness ([N/m]/kg)	74.8 ± 30.2	67.8 ± 14.9	93.7 ± 15.8	97.9 ± 16.8	145.2 ± 34.7
	Force at zero velocity (N/kg)	20.0 ± 2.4	22.2 ± 2.1	22.2 ± 1.6	25.5 ± 2.3	26.4 ± 2.5
	V_{max}-TOV percent difference (%)	7.5 ± 1.8	4.8 ± 0.8	5.4 ± 1.2	4.6 ± 0.8	4.6 ± 1.2
CMJ plus additional external load (60% body mass)	Jump height (cm)	14.4 ± 3.8	23.4 ± 2.5	21.7 ± 3.2	25.7 ± 3.9	24.8 ± 5.1
	Peak propulsive power (W/kg)	46.7 ± 6.3	60.1 ± 4.4	59.5 ± 4.7	63.7 ± 6.5	64.8 ± 8.1
	Rate of power development (W/kg/s)	146.4 ± 37.3	178.9 ± 24.7	217.1 ± 33.0	219.8 ± 30.9	285.9 ± 48.6
	Net eccentric phase impulse (Ns/kg)	1.44 ± 0.24	1.86 ± 0.26	1.68 ± 0.23	2.10 ± 0.20	1.82 ± 0.36
	Peak braking power (W/kg)	16.2 ± 3.7	21.6 ± 4.0	19.8 ± 3.7	26.5 ± 3.8	23.2 ± 5.9
	Concentric phase duration (s)	0.43 ± 0.07	0.42 ± 0.04	0.36 ± 0.04	0.37 ± 0.03	0.31 ± 0.04
	Eccentric phase duration (s)	0.29 ± 0.07	0.29 ± 0.04	0.25 ± 0.03	0.24 ± 0.03	0.20 ± 0.03
	Contraction time (s)	1.19 ± 0.18	1.21 ± 0.10	1.08 ± 0.11	1.10 ± 0.09	0.94 ± 0.13
	Whole body eccentric stiffness ([N/m]/kg)	57.8 ± 24.9	53.0 ± 12.9	73.9 ± 16.3	73.2 ± 14.8	106.5 ± 25.6
	Force at zero velocity (N/kg)	24.7 ± 2.3	26.6 ± 1.9	27.3 ± 1.6	29.5 ± 1.6	30.9 ± 2.1
	V_{max}-TOV percent difference (%)	15.6 ± 4.8	9.4 ± 1.5	10.4 ± 2.3	8.4 ± 2.0	8.6 ± 2.4
Single-leg, CMJ	Jump height (cm)	12.3 ± 3.2	20.0 ± 2.1	19.0 ± 2.0	21.9 ± 3.7	21.8 ± 3.6
	Peak propulsive power (W/kg)	27.0 ± 3.7	34.8 ± 2.7	35.2 ± 2.4	36.8 ± 4.3	38.4 ± 4.3
	Rate of power development (W/kg/s)	111.1 ± 27.0	128.9 ± 18.4	170.7 ± 28.2	157.1 ± 27.8	216.2 ± 38.1
	Net eccentric phase impulse (Ns/kg)	0.74 ± 0.13	0.96 ± 0.13	0.83 ± 0.14	1.15 ± 0.12	0.97 ± 0.17
	Peak braking power (W/kg)	8.4 ± 1.9	11.2 ± 1.8	9.8 ± 1.9	14.9 ± 2.4	12.7 ± 3.1
	Concentric phase duration (s)	0.37 ± 0.05	0.38 ± 0.03	0.31 ± 0.03	0.34 ± 0.03	0.28 ± 0.03
	Eccentric phase duration (s)	0.23 ± 0.04	0.24 ± 0.03	0.19 ± 0.03	0.20 ± 0.03	0.16 ± 0.03
	Contraction time (s)	1.05 ± 0.12	1.11 ± 0.08	0.93 ± 0.10	1.02 ± 0.10	0.87 ± 0.10
	Whole body eccentric stiffness ([N/m]/kg)	69.4 ± 27.6	55.7 ± 18.5	93.4 ± 31.7	77.2 ± 17.7	138.8 ± 56.8
	Force at zero velocity (N/kg)	15.9 ± 1.5	16.7 ± 0.9	17.5 ± 1.0	19.2 ± 1.0	20.1 ± 1.3
	V_{max}-TOV percent difference (%)	19.8 ± 7.4	11.2 ± 2.5	12.0 ± 3.7	10.2 ± 3.3	10.2 ± 3.7
Single-leg repeated hop	Jump height (cm)	7.6 ± 3.9	13.1 ± 3.9	11.0 ± 4.4	16.1 ± 4.6	13.7 ± 4.1
	Contact time (s)	0.37 ± 0.06	0.32 ± 0.06	0.30 ± 0.04	0.33 ± 0.06	0.29 ± 0.04
	Peak ground reaction force (N/kg)	30.3 ± 4.2	36.0 ± 4.5	36.4 ± 4.4	35.5 ± 3.9	39.3 ± 4.5
	Whole body eccentric stiffness ([N/m]/kg)	214.9 ± 102.0	227.6 ± 101.0	282.0 ± 116.4	196.2 ± 101.5	262.2 ± 80.5
Maximal multijoint leg press strength	Concentric MVC (N/kg)	21.2 ± 3.7	23.6 ± 3.6	22.9 ± 3.2	26.0 ± 2.5	26.7 ± 3.7
	Eccentric MVC (N/kg)	22.3 ± 5.2	26.5 ± 4.8	24.7 ± 4.8	28.8 ± 4.7	29.2 ± 5.1

*Cluster 1 ($Cluster_{LOW}$) displayed low body mass normalized muscle strength and power, cluster 2 ($Cluster_{SLOW}$) displayed average muscle strength and slow jump contraction time, cluster 3 ($Cluster_{AVG}$) displayed average muscle strength and average jump contraction time, cluster 4 ($Cluster_{STRONG-BRAKE}$) displayed high muscle strength and power, as well as a braking-dominant jump strategy, cluster 5 ($Cluster_{STRONG-FAST}$) displayed high muscle strength and power, along with fast jump contraction time.

CMJ = 29.3 ± 5.1 cm, CMJ_{loaded} = 14.4 ± 3.8 cm, CMJ_{SL} = 12.3 ± 3.2 cm; peak propulsive power: CMJ = 46.3 ± 5.5 W/kg, CMJ_{loaded} = 46.7 ± 6.3 W/kg, CMJ_{SL} = 27.0 ± 3.2 W/kg), lowest net eccentric deceleration (braking) impulse (CMJ = 1.08 ± 0.18 Ns/kg, CMJ_{loaded} = 1.44 ± 0.24 Ns/kg, CMJ_{SL} = 0.74 ± 0.13 Ns/kg), and longer than average concentric jump phase duration (CMJ = 0.30 ± 0.04 seconds). In addition, athletes in $Cluster_{LOW}$ were characterized by the largest V_{max}-TOV percent difference (CMJ = 7.5 ± 1.8%) and the lowest MVC force during both concentric (21.2 ± 3.7 N/kg) and eccentric (22.3 ± 5.2 N/kg) actions (Table 1).

Cluster 2: Average Strength and Power, Slow Jump Contraction Time ($Cluster_{SLOW}$)

$Cluster_{SLOW}$ (n = 65; 23.9% of the total sample) was made up of tackle football (32.3% of the cluster), soccer (27.7% of the cluster), volleyball (18.5% of the cluster), ice hockey (12.3% of the cluster), and basketball (9.2% of the cluster) athletes. This was equivalent to 13.2% of the tackle football population, 40.9% of the soccer population, 41.5% of the volleyball population, 34.8% of the ice hockey population, and 35.2% of the basketball population. Biomechanics of this group (Figure 3) were

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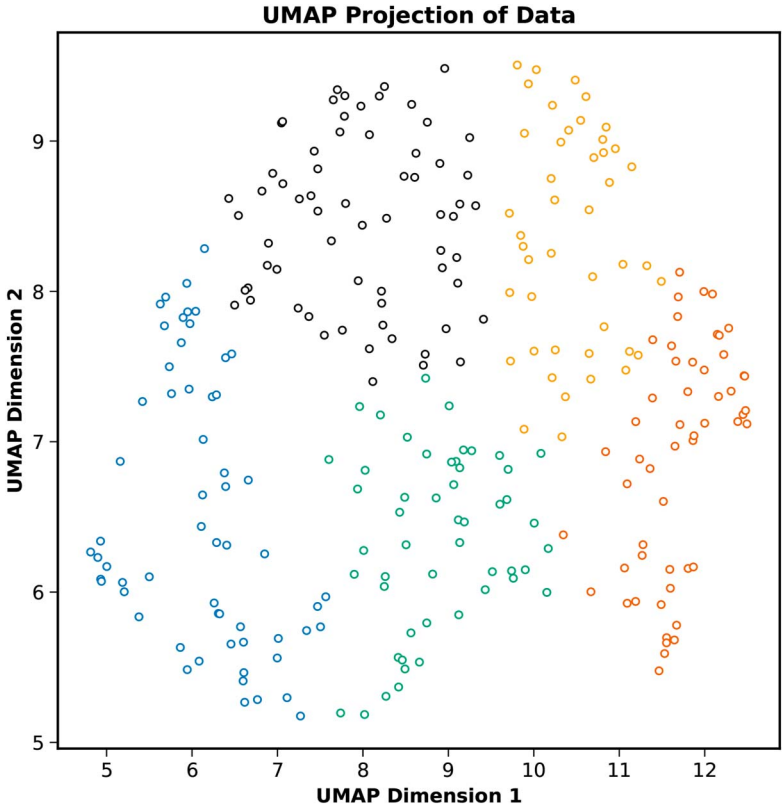


Figure 1. Two-dimensional visual representation of clustering assignments using Uniform Manifold Approximation and Projection for Dimension Reduction (UMAP). Colors are representative of different athlete clusters.

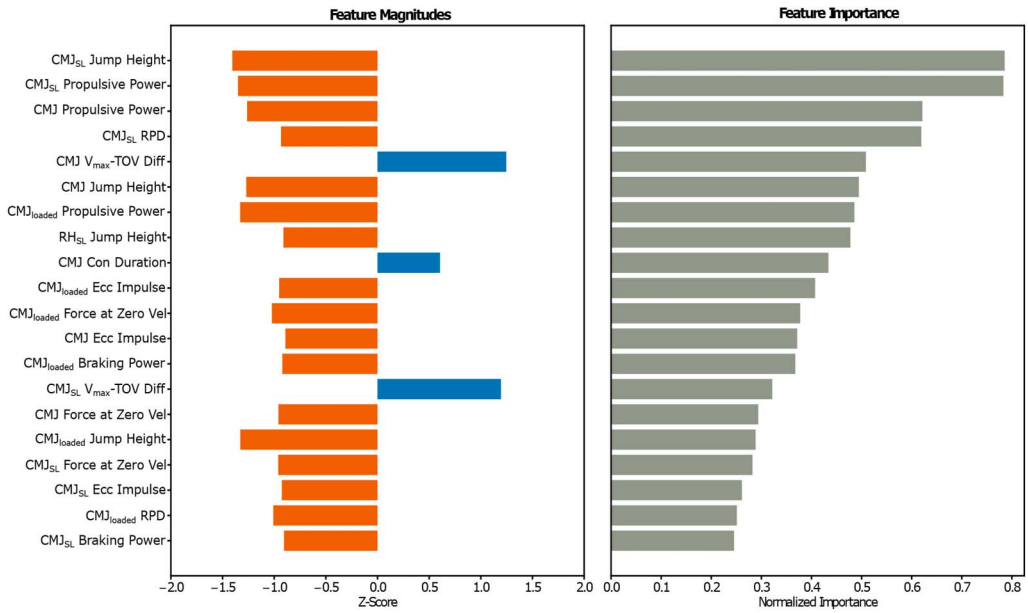


Figure 2. The average feature importance for the top 20 most important features in assigning athletes to $Cluster_{LOW}$. CMJ: body weight countermovement jump; CMJ_{SL}: single-leg body weight countermovement jump; CMJ_{loaded}: countermovement jump plus additional external load (60% body mass); RH_{SL}: single-leg repeated hop; CON_{MVC}: concentric multijoint maximal voluntary contraction; ECC_{MVC}: eccentric multijoint maximal voluntary contraction; Con: concentric; Ecc: eccentric; LL_{STIFF}: lower-body eccentric deceleration stiffness; RPD: rate of propulsive power development; V_{max}-TOV Diff: percent difference between the maximal velocity obtained during the concentric phase (V_{max}) and the velocity at takeoff (TOV); concentric and eccentric impulses are calculated as the net concentric and eccentric deceleration impulse.

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characteristic of athletes with the longest CMJ concentric (CMJ = 0.31 ± 0.03 seconds, CMJ_{loaded} = 0.42 ± 0.04 seconds, CMJ_{SL} = 0.38 ± 0.03 seconds) and CMJ eccentric deceleration (CMJ = 0.21 ± 0.03 s, CMJ_{loaded} = 0.29 ± 0.04 seconds, CMJ_{SL} = 0.24 ± 0.03 seconds) phase contraction times. In addition, this group demonstrated an average or above-average CMJ net eccentric deceleration impulse (CMJ = 1.36 ± 0.17 Ns/kg, CMJ_{loaded} = 1.86 ± 0.26 Ns/kg, CMJ_{SL} = 0.96 ± 0.13 Ns/kg) and CMJ propulsive performance (jump height: CMJ = 41.3 ± 3.5 cm, CMJ_{loaded} = 23.4 ± 2.5 cm, CMJ_{SL} = 20.0 ± 2.1 cm) (Table 1).

Cluster 3: Average Muscle Strength and Power, Average Jump Contraction Time ($Cluster_{AVG}$)

$Cluster_{AVG}$ (n = 51; 18.8%) was made up of tackle football (54.9% of the cluster), soccer (19.6% of the cluster), basketball (13.7% of the cluster), and volleyball (11.8% of the cluster) athletes. This was equivalent to 17.5% of the tackle football population, 22.7% of the soccer population, 41.1% of the basketball population, 20.8% of the volleyball population, and 0.0% of the ice hockey population. Biomechanical features of $Cluster_{AVG}$ are shown in Figure 4. Athletes in $Cluster_{AVG}$ demonstrated average or below-average CMJ performance (net eccentric deceleration impulse: CMJ = 1.18 ± 0.15 Ns/kg, CMJ_{loaded} = 1.68 ± 0.23 Ns/kg, CMJ_{SL} = 0.83 ± 0.14 Ns/kg; jump height: CMJ = 37.1 ± 4.1 cm, CMJ_{loaded} = 21.7 ± 3.2 cm, CMJ_{SL} = 19.0 ± 2.0 cm) and duration (eccentric deceleration duration: CMJ = 0.18 ± 0.02 seconds, CMJ_{loaded} = 0.25 ± 0.03 seconds, CMJ_{SL} = 0.19 ± 0.03 seconds; concentric duration: CMJ = 0.26 ± 0.02 seconds, CMJ_{loaded} = 0.36 ± 0.04 seconds, CMJ_{SL} = 0.31 ± 0.03 seconds), along with below-average maximal lower-body muscle strength (CON_{MVC} = 22.9 ± 3.2 N/kg; ECC_{MVC} = 24.7 ± 4.8 N/kg) (Table 1).

Cluster 4: High Strength and Power, Braking-Dominant Strategy ($Cluster_{STRONG\text{-}BRAKE}$)

$Cluster_{STRONG\text{-}BRAKE}$ (n = 44; 16.2% of total sample) was made up of tackle football (50.0% of the cluster), ice hockey (29.5% of the cluster), volleyball (15.9% of the cluster), basketball (2.3% of the cluster), and soccer (2.3% of the cluster) athletes. This was equivalent to 13.8% of the tackle football population, 56.5% of the ice hockey population, 24.1% of the volleyball population, 6.0% of the basketball population, and 2.3% of the soccer population. Biomechanical features of $Cluster_{STRONG\text{-}BRAKE}$ are shown in Figure 5. Athletes in $Cluster_{STRONG\text{-}BRAKE}$ exhibited the highest eccentric deceleration force output (net eccentric deceleration impulse: CMJ = 1.55 ± 0.15 Ns/kg, CMJ_{loaded} = 2.10 ± 0.20 Ns/kg, CMJ_{SL} = 1.15 ± 0.12 Ns/kg; peak braking power: CMJ = 23.4 ± 3.7 W/kg, CMJ_{loaded} = 26.5 ± 3.8 W/kg, CMJ_{SL} = 14.9 ± 2.4 W/kg) and average eccentric deceleration contraction durations (CMJ = 0.17 ± 0.02 seconds, CMJ_{loaded} = 0.24 ± 0.03 seconds, CMJ_{SL} = 0.20 ± 0.03 seconds). In addition, this group displayed the highest jump performance (jump height: CMJ = 44.0 ± 5.3 cm, CMJ_{loaded} = 25.7 ± 3.9 cm, CMJ_{SL} = 21.9 ± 3.7 cm), as well as above-average concentric (26.0 ± 2.5 N/kg) and eccentric (28.8 ± 4.7 N/kg) maximal muscle strength (Table 1).

Cluster 5: High Strength and Power, Fast Jump Contraction Time ($Cluster_{STRONG\text{-}FAST}$)

$Cluster_{STRONG\text{-}FAST}$ (n = 53; 19.5% of the total sample) was made up of tackle football (79.2% of the cluster), soccer (17.0% of the cluster), ice hockey (1.9% of the cluster), and basketball (1.9% of the cluster) athletes. This was equivalent to 26.4% of the football population, 20.5% of the soccer population, 4.4% of the ice hockey population, 5.9% of the basketball population, and

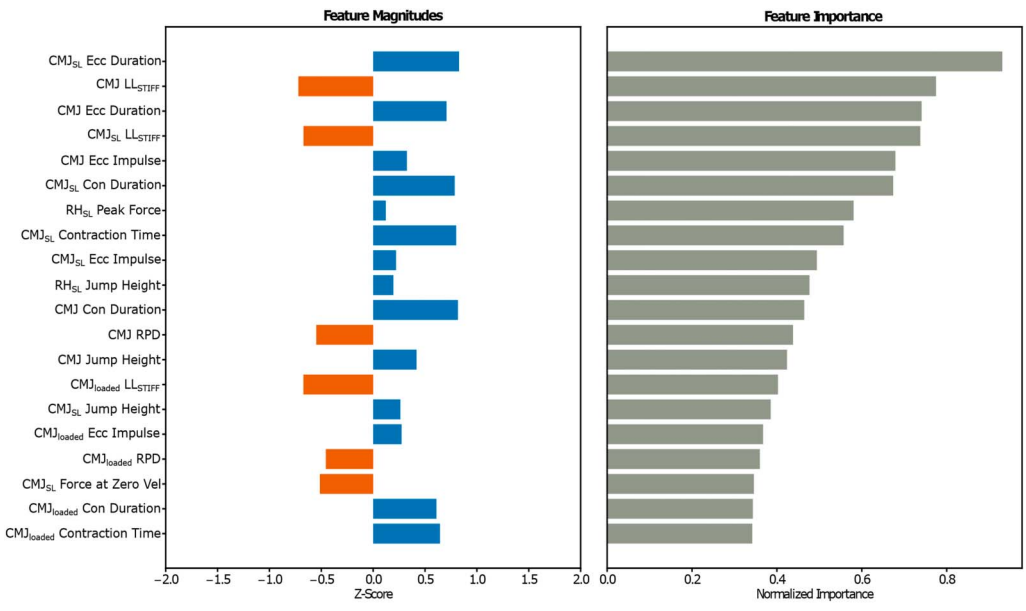


Figure 3. The average feature importance for the top 20 most important features in assigning athletes to $Cluster_{SLOW}$. CMJ: bodyweight countermovement jump; CMJ_{SL}: single-leg countermovement jump; CMJ_{loaded}: countermovement jump plus additional external load (60% body mass); RH_{SL}: single-leg repeated hop; CON_{MVC}: concentric multijoint maximal voluntary contraction; ECC_{MVC}: eccentric multijoint maximal voluntary contraction; Con: concentric; Ecc: eccentric; LL_{STIFF}: lower-body eccentric deceleration stiffness; RPD: rate of propulsive power development; V_{max}-TOV Diff: percent difference between the maximal velocity obtained during the concentric phase (V_{max}) and the velocity at takeoff (TOV); concentric and eccentric impulses are calculated as the net concentric and eccentric deceleration impulse.

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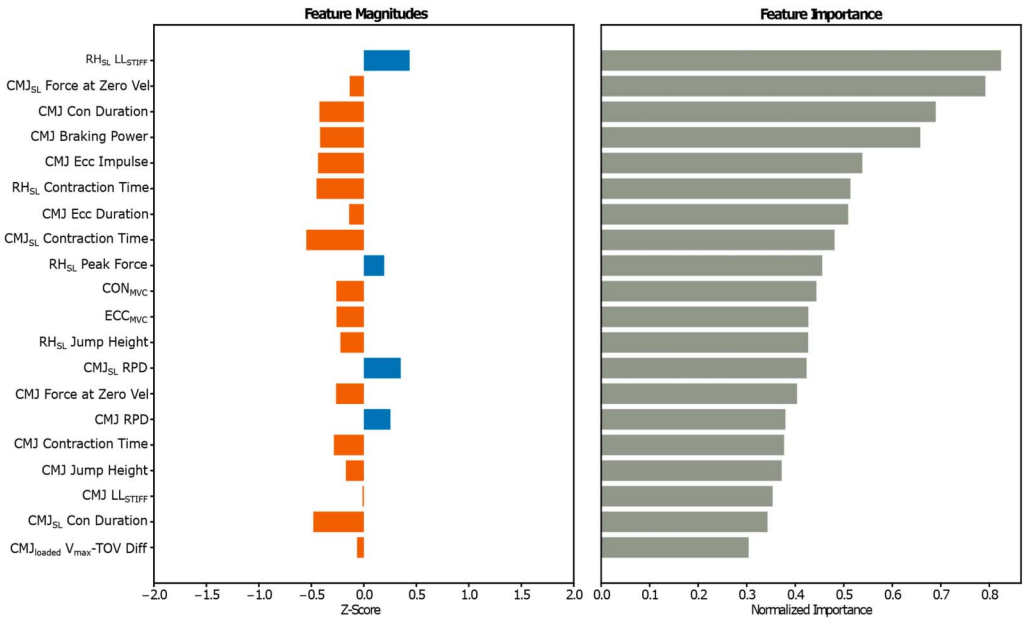


Figure 4. The average feature importance for the top 20 most important features in assigning athletes to $Cluster_{AVG}$. CMJ: bodyweight countermovement jump; CMJ_{SL}: single-leg countermovement jump; CMJ_{loaded}: countermovement jump plus additional external load (60% body mass); RH_{SL}: single-leg repeated hop; CON_{MVC}: concentric multijoint maximal voluntary contraction; ECC_{MVC}: eccentric multijoint maximal voluntary contraction; Con: concentric; Ecc: eccentric; LL_{STIFF}: lower-body eccentric deceleration stiffness; RPD: rate of propulsive power development; V_{max}-TOV Diff: percent difference between the maximal velocity obtained during the concentric phase (V_{max}) and the velocity at takeoff (TOV); concentric and eccentric impulses are calculated as the net concentric and eccentric deceleration impulse.

0.0% of the volleyball population. Biomechanical characteristics (Figure 6) show that athletes in $Cluster_{STRONG\text{-}FAST}$ exhibited the best jump performance (jump height: CMJ = 41.3 ± 6.4 cm, CMJ_{loaded} = 24.8 ± 5.1 cm, CMJ_{SL} = 21.8 ± 3.6 cm; peak propulsive power: CMJ = 62.3 ± 6.9 W/kg, CMJ_{loaded} = 64.8 ± 8.1 W/kg, CMJ_{SL} = 38.4 ± 4.3 W/kg), along with the shortest

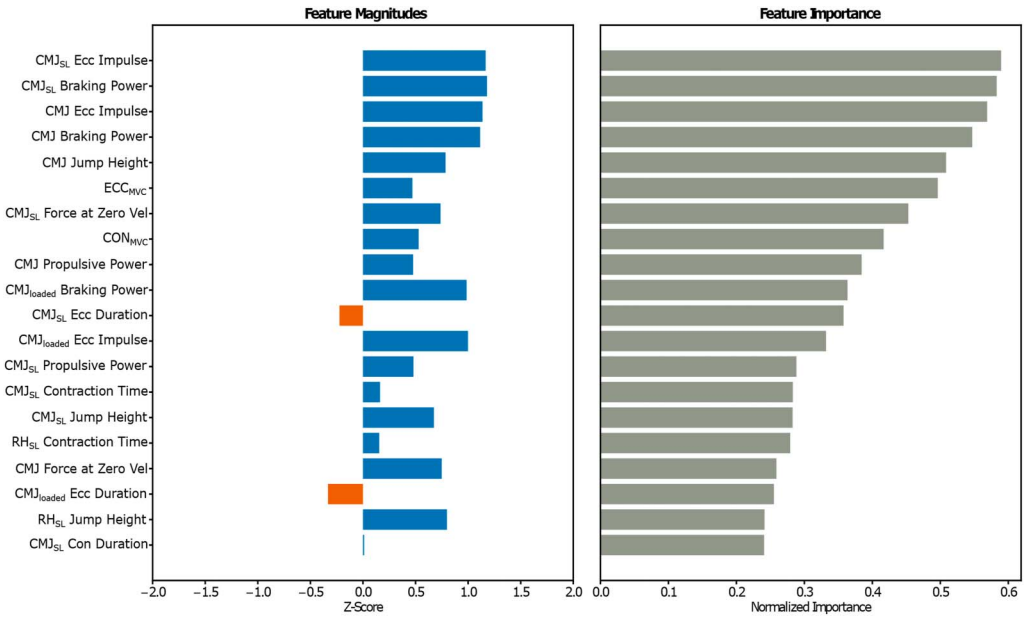


Figure 5. The average feature importance for the top 20 most important features in assigning athletes to $Cluster_{STRONG\text{-}BRAKE}$. CMJ: body weight countermovement jump; CMJ_{SL}: single-leg body weight countermovement jump; CMJ_{loaded}: countermovement jump plus additional external load (60% body weight); RH_{SL}: single-leg repeated hop; CON_{MVC}: concentric multijoint maximal voluntary contraction; ECC_{MVC}: eccentric multijoint maximal voluntary contraction; Con: concentric; Ecc: eccentric; LL_{STIFF}: lower-body eccentric deceleration stiffness; RPD: rate of propulsive power development; V_{max}-TOV Diff: percent difference between the maximal velocity obtained during the concentric phase (V_{max}) and the velocity at takeoff (TOV); concentric and eccentric impulses are calculated as the net concentric and eccentric deceleration impulse.

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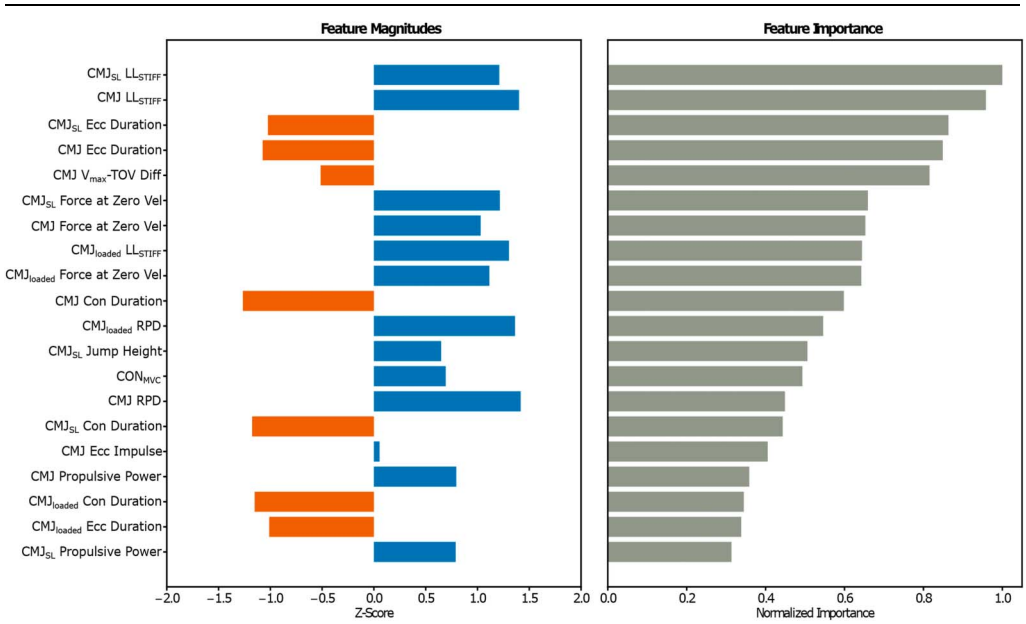


Figure 6. The average feature importance for the top 20 most important features in assigning athletes to Cluster$_{STRONG-FAST}$. CMJ: bodyweight countermovement jump; CMJ$_{SL}$: single-leg ountermovement jump; CMJ$_{loaded}$: countermovement jump plus additional external load (60% body mass); RH$_{SL}$: single-leg repeated hop; CON$_{MVC}$: concentric multijoint maximal voluntary contraction; ECC$_{MVC}$: eccentric multijoint maximal voluntary contraction; Con: concentric; Ecc: eccentric; LL$_{STIFF}$: lower-body eccentric deceleration stiffness; RPD: rate of propulsive power development; V$_{max}$-TOV Diff: percent difference between the maximal velocity obtained during the concentric phase (V$_{max}$) and the velocity at takeoff (TOV); concentric and eccentric impulses are calculated as the net concentric and eccentric deceleration impulse.

CMJ concentric (CMJ = 0.23 ± 0.02 seconds) and eccentric deceleration (CMJ = 0.14 ± 0.02 seconds) phase durations. In addition, the highest eccentric deceleration lower limb stiffness (CMJ = 145.2 ± 34.7 (N/cm)/kg) and rate of propulsive power development (CMJ = 377.3 ± 59.1 W/s/kg) were identified as highly important features in the assignment of athletes to Cluster$_{STRONG-FAST}$ (Figure 6). Cluster$_{STRONG-FAST}$ also displayed the highest concentric (26.7 ± 3.7 N/kg) and eccentric (29.2 ± 5.1 N/kg) maximal muscle force production capacity.

Biopsychosocial and Injury Characteristics of the Clusters

An effect of body mass ($F_{4,267} = 48.6$, $p < 0.001$, $R^2 = 0.42$) on the athlete cluster was observed, where athletes in Cluster$_{LOW}$ (115.8 ± 25.9 kg) had a greater body mass compared with all other clusters ($p < 0.001$). An effect of age ($F_{4,267} = 3.86$, $p = 0.005$, $R^2 = 0.055$) on the athlete cluster was also noted, where athletes in Cluster$_{STRONG-BRAKE}$ (21.6 ± 1.9 years) were older than all other clusters ($p = 0.002$). No effect of experience level (i.e., number of years competing in U Sport; $p = 0.38$) or perception of overall MSK function (i.e., SFI score; $p = 0.59$) on cluster assignment was observed.

An effect of athlete attitudes toward strength training and injury prevention on clustering assignment was observed (Table 2; questions with significant differences shown in Figure 7). Of note, Cluster$_{LOW}$ included more athletes who stated a preference for strength training in a private setting rather than in a public setting (Figure 7D, $p = 0.023$). In addition, different responses across clusters were observed for questions related to the risk of injury in their sport (Figure 7A, B; $p < 0.051$) and the importance of including injury prevention programs within a holistic athlete training program (Figure 7C–F; $p < 0.037$). No differences across clusters for any question regarding athlete attitudes and beliefs toward training were observed (Table 2).

Discussion

This study identified 5 neuromuscular performance profiles in healthy male university athletes by applying an unsupervised ML clustering analysis to a multifaceted dataset that was obtained through a biomechanical force analysis of double-leg CMJ testing with and without extra load, single-leg CMJ testing, single-leg repeat hop testing, and multijoint leg extension testing on a robotic leg press (23). While these biomechanical force assessments are becoming increasingly popular for athlete testing (2,4), they yield an extensive array of outcome measures that have inherent challenges when interpreting these multifaceted datasets. Unsupervised ML algorithms can be used to characterize relevant athlete subpopulations by identifying athletes with similar profiles based on a weighted pattern of neuromuscular performance capacities across multiple outcome metrics (5). In addition, the testing outcome metrics that most prominently distinguish these profiles can be identified (19). Together, this analysis strategy may be useful for establishing key performance indicators and neuromuscular benchmarks for preseason performance testing or establishing performance-readiness after injury.

The unsupervised ML approach used in our study identified 5 neuromuscular performance profiles that included: Cluster$_{LOW}$ (22% of the athletes showing low body mass normalized strength and power, and low jump height); Cluster$_{SLOW}$ (24% of the athletes showing average body mass normalized strength and power, average jump height, and the slowest jump contraction times); Cluster$_{AVG}$ (19% of the athletes showing average body mass normalized strength and power, and average jump contraction times); Cluster$_{STRONG-BRAKE}$ (16% of the athletes showing high body mass normalized strength and power, and a braking-dominant jump movement strategy); and Cluster$_{STRONG-FAST}$ (20% of the athletes showing high body mass normalized strength and power,

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Table 2

Responses to the Attitudes Toward Strength Training and Injury Prevention Questionnaire adapted from MØller et al. (28).*

Attitudes and beliefs toward non–sport-specific training question	Cluster	Number of responses							Fisher's exact test *p*
		Strongly disagree	Disagree	Slightly disagree	Neither	Slightly agree	Agree	Strongly agree	
"The risk of injury is high in my sport"	$Cluster_{LOW}$	0	2	1	3	5	24	24	0.002
	$Cluster_{SLOW}$	2	0	2	2	16	23	18	
	$Cluster_{AVG}$	0	0	3	1	11	16	17	
	$Cluster_{STRONG-BRAKE}$	1	0	1	0	4	21	16	
	$Cluster_{STRONG-FAST}$	0	0	0	2	2	25	22	
"The risk of anterior cruciate ligament (ACL) injury is high in my sport"	$Cluster_{LOW}$	0	0	1	3	6	22	26	0.051
	$Cluster_{SLOW}$	2	0	1	7	8	32	13	
	$Cluster_{AVG}$	0	2	3	2	9	22	10	
	$Cluster_{STRONG-BRAKE}$	1	3	1	1	4	19	14	
	$Cluster_{STRONG-FAST}$	0	0	0	4	3	24	20	
"Many sport injuries are preventable"	$Cluster_{LOW}$	0	0	1	12	19	22	4	0.976
	$Cluster_{SLOW}$	0	1	3	7	24	23	5	
	$Cluster_{AVG}$	0	1	3	6	16	18	4	
	$Cluster_{STRONG-BRAKE}$	0	0	1	4	21	14	3	
	$Cluster_{STRONG-FAST}$	0	1	5	4	17	18	6	
"ACL injury prevention is important"	$Cluster_{LOW}$	1	0	0	1	2	19	35	0.037
	$Cluster_{SLOW}$	0	0	0	0	1	20	42	
	$Cluster_{AVG}$	0	0	0	0	0	16	32	
	$Cluster_{STRONG-BRAKE}$	0	0	0	0	4	15	24	
	$Cluster_{STRONG-FAST}$	0	0	0	0	0	20	31	
"Injury prevention will be a priority for me this upcoming season"	$Cluster_{LOW}$	0	0	0	4	4	26	24	0.561
	$Cluster_{SLOW}$	0	0	0	0	4	20	39	
	$Cluster_{AVG}$	0	0	0	0	5	15	28	
	$Cluster_{STRONG-BRAKE}$	0	0	0	1	2	16	24	
	$Cluster_{STRONG-FAST}$	0	0	0	0	3	20	28	
"It is important to have a consistent weight training program each week"	$Cluster_{LOW}$	0	0	0	3	1	29	25	0.564
	$Cluster_{SLOW}$	0	1	0	0	2	29	31	
	$Cluster_{AVG}$	0	0	0	0	5	24	19	
	$Cluster_{STRONG-BRAKE}$	0	0	0	0	4	17	22	
	$Cluster_{STRONG-FAST}$	1	0	0	0	2	24	24	
"Focusing on aerobic exercise (instead of strength training) produces greater benefits to health and body appearance"	$Cluster_{LOW}$	1	4	13	16	12	9	3	0.438
	$Cluster_{SLOW}$	0	10	5	20	10	12	6	
	$Cluster_{AVG}$	1	1	5	17	14	8	2	
	$Cluster_{STRONG-BRAKE}$	0	5	5	17	6	7	3	
	$Cluster_{STRONG-FAST}$	0	3	5	18	10	13	2	
"Strength training in a private setting (or at my own home) is more comfortable than a gym setting"	$Cluster_{LOW}$	4	14	5	15	7	8	5	0.023
	$Cluster_{SLOW}$	1	23	2	28	6	0	3	
	$Cluster_{AVG}$	0	14	4	20	3	7	0	
	$Cluster_{STRONG-BRAKE}$	3	11	6	17	2	3	1	
	$Cluster_{STRONG-FAST}$	1	6	7	20	11	4	2	
"Exercise that reduces overall body weight is more important to me than increasing lean muscle mass"	$Cluster_{LOW}$	9	16	7	14	6	2	4	0.271
	$Cluster_{SLOW}$	10	28	12	9	0	0	4	
	$Cluster_{AVG}$	11	9	9	13	2	4	0	
		13	16	7	5	0	2	0	

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Table 2

Responses to the Attitudes Toward Strength Training and Injury Prevention Questionnaire adapted from MØller et al. (28).* *(Continued)*

Attitudes and beliefs toward non–sport-specific training question	Cluster	Number of responses							Fisher's exact test *p*
		Strongly disagree	Disagree	Slightly disagree	Neither	Slightly agree	Agree	Strongly agree	
	$Cluster_{STRONG-BRAKE}$								
	$Cluster_{STRONG-FAST}$	14	13	8	8	4	3	1	
"Strength training is more enjoyable when only lifting my body weight or light/easy weights (versus heavy weights)"	$Cluster_{LOW}$	13	15	9	13	3	4	1	0.286
	$Cluster_{SLOW}$	12	16	13	10	7	4	1	
	$Cluster_{AVG}$	7	19	9	4	6	3	1	
	$Cluster_{STRONG-BRAKE}$	10	19	8	4	2	0	0	
	$Cluster_{STRONG-FAST}$	7	17	6	13	3	5	0	
"It is more important to use training time to play your sport (i.e., basketball, soccer) rather than implement program designed to reduce injury"	$Cluster_{LOW}$	1	12	9	15	13	7	1	0.067
	$Cluster_{SLOW}$	3	15	14	14	11	4	2	
	$Cluster_{AVG}$	3	13	13	10	3	5	1	
	$Cluster_{STRONG-BRAKE}$	2	9	10	8	9	4	1	
	$Cluster_{STRONG-FAST}$	1	12	3	13	5	15	2	
"Training should focus more on improving game play rather than reducing injury"	$Cluster_{LOW}$	1	3	8	21	19	5	1	0.026
	$Cluster_{SLOW}$	5	9	9	10	16	14	1	
	$Cluster_{AVG}$	1	4	13	12	11	5	1	
	$Cluster_{STRONG-BRAKE}$	2	6	10	9	11	4	1	
	$Cluster_{STRONG-FAST}$	0	4	12	8	13	13	1	
"I would do extra training if it would reduce my injury risk"	$Cluster_{LOW}$	1	0	3	3	19	21	11	0.016
	$Cluster_{SLOW}$	0	0	0	3	16	32	12	
	$Cluster_{AVG}$	0	1	0	3	14	16	13	
	$Cluster_{STRONG-BRAKE}$	0	0	1	1	5	24	12	
	$Cluster_{STRONG-FAST}$	0	0	0	3	7	29	12	
"Players who do not partake in injury prevention training are more at risk of suffering an injury"	$Cluster_{LOW}$	0	0	0	2	18	17	21	0.189
	$Cluster_{SLOW}$	0	1	1	3	5	26	17	
	$Cluster_{AVG}$	0	0	0	3	7	28	9	
	$Cluster_{STRONG-BRAKE}$	0	0	1	0	10	19	13	
	$Cluster_{STRONG-FAST}$	1	0	1	5	6	23	15	

*Cluster 1 ($Cluster_{LOW}$) displayed low body weight–normalized muscle strength and power, cluster 2 ($Cluster_{SLOW}$) displayed average muscle strength and slow jump contraction time, cluster 3 ($Cluster_{AVG}$) displayed average muscle strength and average jump contraction time, cluster 4 ($Cluster_{STRONG-BRAKE}$) displayed high muscle strength and power, as well as a braking-dominant jump strategy, cluster 5 ($Cluster_{STRONG-FAST}$) displayed high muscle strength and power, along with fast jump contraction time.

and fast jump contraction times). There was evidence of sport-specificity with $Cluster_{LOW}$ including a majority of offensive linemen and linebackers, $Cluster_{STRONG-BRAKE}$ including most ice hockey players, and $Cluster_{STRONG-FAST}$ with high representation from field-sport sprinting athletes. There were 2 ACL injuries over the course of this study, 1 occurred in $Cluster_{STRONG-FAST,}$ and 1 occurred in $Cluster_{LOW}$. While the injury surveillance aspect of this data collection did not produce a sufficient number of injuries to provide actionable insights, applying a complex systems approach for the identification of injury risk patterns has been encouraged previously (5), and our study provides an analytical framework for further research in this area.

In addition to highlighting the methodological application of unsupervised ML to neuromuscular performance datasets, in alignment with research groups in other areas (7,30), the results of this study may also be used to inform key performance indicators for a healthy male university athlete population. Our identification of the biomechanical force testing outcome metrics that most prominently distinguish each neuromuscular performance profile could be used by practitioners to establish data-driven key performance indicators (see Figures 2–6) and an associated minimum performance target for each key performance indicator (see Table 1). This can be used to provide more specific context for the interpretation of preseason baseline testing or for establishing neuromuscular performance benchmarks that can be used as an alternative to the contralateral limb benchmark when guiding rehabilitation after injury (39). Although this study may not be generalizable to other populations of male elite athletes, this study illuminates opportunities for advancements in the application of sport science research in settings where a broad battery of biomechanical assessments is used to evaluate athlete neuromuscular performance capacities, such as in professional sport, the military, or clinical environments.

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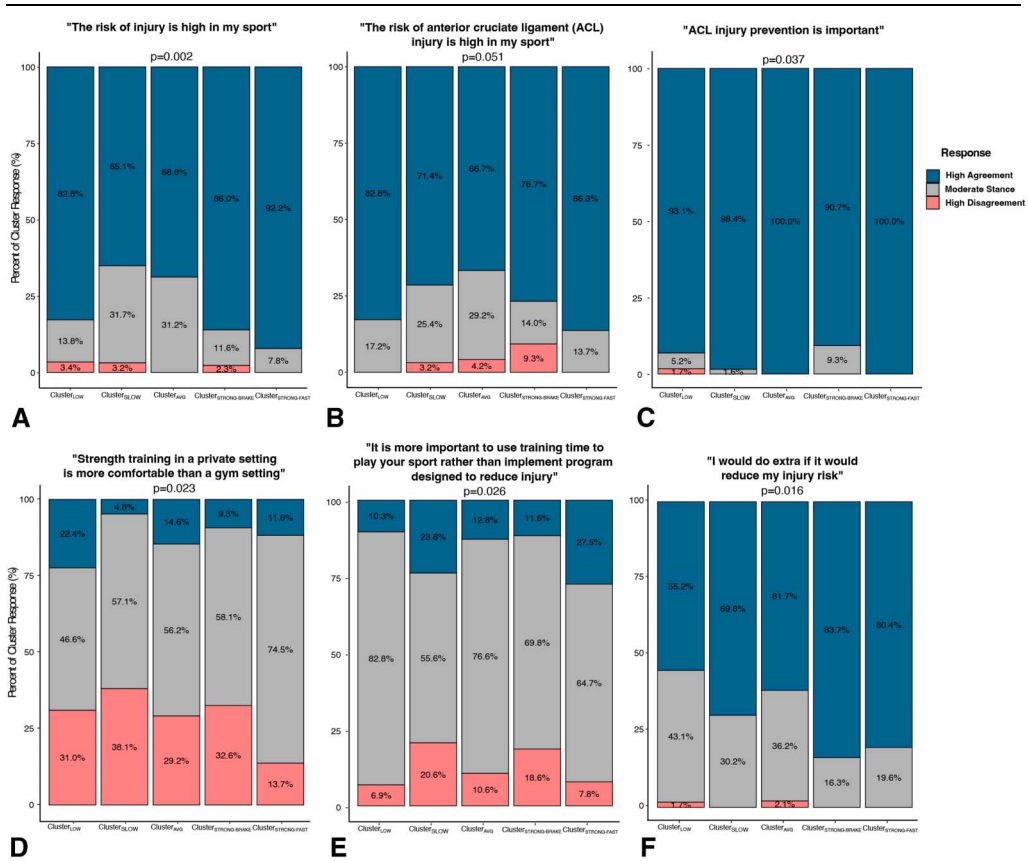


Figure 7. Summary of the Attitudes Toward Strength Training and Injury Prevention Questionnaire (adapted from Møller et al. (27)) Athlete responses across clusters for the questions "The risk of injury is high in my sport' (A), "The risk of anterior cruciate ligament (ACL) injury is high in my sport" (B), "ACL injury prevention is important" (C), "Strength training in a private setting is more comfortable than in a gym setting" (D), "It is more important to use training time to play your sport rather than implemnt injury programs designed to reduce injury" (E), and "I would do extra if it would reduce my injury risk" (F) are shown. Cluster 1 ($Cluster_{LOW}$) displayed low body mass normalized muscle strength and power, cluster 2 ($Cluster_{SLOW}$) displayed average muscle strength and slow jump contraction times, cluster 3 ($Cluster_{AVG}$) displayed average muscle strength and average jump contraction times, cluster 4 ($Cluster_{STRONG\text{-}BRAKE}$) displayed high muscle strength and power, as well as a braking-dominant movement strategy, and cluster 5 ($Cluster_{STRONG\text{-}FAST}$) displayed high muscle strength and power, as well as fast jump contraction times. To isolate subjects with more extreme responses, the 7-point Likert scale was categorized into 3 broader groups: (a) strongly agree and agree responses were grouped to reflect high agreement with the statement; (b) slightly agree, neither, and slightly disagree responses were grouped to reflect a moderate stance toward the statement; and (c) strongly disagree and disagree responses were grouped to reflect high disagreement with the statement. Fisher's exact test was computed to test the effect of cluster assignment on attitudes toward strength training and injury prevention.

A primary difference in this analysis of healthy male university athletes compared with that of healthy female university athletes (25) is the evidence of sport-specificity across neuromuscular performance profiles. For instance, $Cluster_{STRONG\text{-}BRAKE}$ contained 56% of the ice hockey players in this study. This neuromuscular performance profile was characterized by a braking-dominant vertical jump strategy with high maximal strength, power, and vertical jump height which is consistent with previous reports describing the neuromuscular capacities of elite skating athletes (1,38). Comparatively, 96% of the $Cluster_{STRONG\text{-}FAST}$ profile comprised field-sport sprinting athletes (e.g., tackle football wide receivers and defensive backs, and soccer athletes) who exhibited a high maximal strength, a fast and high CMJ strategy, and increased lower limb stiffness, which are associated with sprint performance (36).

Evidence of position specificity was also found whereby, in comparison with the football sprinting athletes who were assigned to $Cluster_{STRONG\text{-}FAST}$, 91.0% of the football players assigned to $Cluster_{LOW}$ were linemen and linebackers. It should be emphasized that this study employed body mass normalization for all outcome measures, which has limitations for interpreting the results specific to high body mass football athletes who possess high absolute strength (34), and future research should consider separate analyses for athletes where absolute strength is the primary driver of performance. However, it is notable that $Cluster_{LOW}$ also had representation from other sports, such as soccer, where athletes demonstrated low absolute and relative strength. These athletes can be identified by practitioners who can implement training programs aimed at the development of neuromuscular performance characteristics that are more representative of a target profile. Future research may consider replicating this study in different sport environments or settings to determine whether strength training practices or sport factors per se are the primary driver of our observations. From a methodological perspective, offshoots of standard unsupervised ML clustering, such as open-set classification, may have utility in the identification of

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new athlete profiles when conducting similar neuromuscular performance evaluations in different contexts (e.g., different sports, aged athletes, performance levels) (26).

Beyond the physiological/biomechanical assessments, this study examined contextual factors related to attitudes toward training and perception of overall musculoskeletal function. While psychosocial factors are often studied with respect to strength training in women (13,24), there is limited research in a male university athlete population. The results from this study demonstrated that a greater proportion of athletes in $Cluster_{LOW}$ preferred performing strength training in a private environment and believed that training time should be used to target sport-specific adaptations rather than for injury prevention (Figure 7). Although the underlying factors influencing the development of these attitudes are unknown, previous literature has suggested that a wide range of factors, such as prevalence of injury in an athlete's sport, sport-specific training culture, body weight, and body image, can modulate strength training behaviors in male university athletes (14,28). There was also evidence of high variability in athlete attitudes toward training and injury prevention both within and between neuromuscular performance profiles. This may highlight the need for an individualized approach when considering the role of psychosocial and environmental factors related to strength training.

Practical Applications

The results from this study have several applications for strength & conditioning coaches, sport scientists, and clinicians. One application of this work is to identify data-driven key performance indicators that are specific to a target athlete profile. Figures 2–6 can be used to identify the most important metrics that differentiate between each neuromuscular performance profile reported in this study. This information can be used by practitioners to first identify an athlete-specific target profile based on their sport/position-specific demands and then establish a set of specific key performance indicators used for preseason performance assessment and in-season monitoring. A second application of this work is that the characteristics of a target profile (Table 1) can be used to establish neuromuscular performance benchmarks. This information can help identify trainable deficits in healthy athletes and guide rehabilitation after lower-body injury like ACL rupture, where new benchmarking approaches are needed (3). A third application of this work is that by combining biomechanical, psychological, and social data streams into a comprehensive analysis of athlete neuromuscular performance, we move toward identifying potential barriers for engagement in heavy strength training on an individual basis. Together, this unsupervised ML approach could provide a framework for future analyses that can be used as part of a practitioner-friendly approach to implementing complex systems–based analyses into applied sport science settings. This study was conducted on a large sample of male university athletes and used an unsupervised ML approach to identify 5 neuromuscular performance profiles with a silhouette score of 0.44. Considering that some overlap between clusters is anticipated in biological datasets, this silhouette score is indicative of a moderately well-defined cluster structure. It is possible that including additional streams of data, such as biomechanical data describing lower-body joint angles during different tasks, could have increased cluster separation. Another important consideration is that we normalized all relevant strength measures to body mass. Although this is a common approach in exercise science, as well as strength and conditioning, it has limitations (29) and, in this case, has contributed to high body mass athletes like tackle football linemen being classified as low-strength athletes. Future research should explore a similar analysis structure using absolute strength capacities, which may provide important additional information to coaches, clinicians, and sport scientists, especially in the context of sport positions that require athletes to have a high body mass. We identified 5 neuromuscular performance profiles that were associated with specific sports environments in male university athletes. Notably, 56% of the ice hockey athletes were profiled as high strength/high power/high braking capacity athletes, and field-sport sprinting athletes (i.e., wide receivers, defensive backs, soccer players) were profiled as high strength/high power/fast muscle contraction/high lower limb stiffness athletes. Differences in athlete attitudes toward strength training and injury prevention were present between profiles, which provides evidence linking biological and psychosocial determinants of neuromuscular expression in this population. These findings have important implications in the construction of neuromuscular performance benchmarks, which are particularly needed to guide rehabilitation and evaluate performance-readiness after lower-body injuries like ACL injury.

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